Vehicle Model in Gazebo - Team 17

Autonomous Systems Project

March 2026

Introduction

This document presents the Gazebo-imported vehicle model for Team 17 and lists the associated ROS 2 topics that can be published and subscribed to for controlling and monitoring the vehicle.

Gazebo Vehicle Screenshots

- **Terminal Screenshots**:

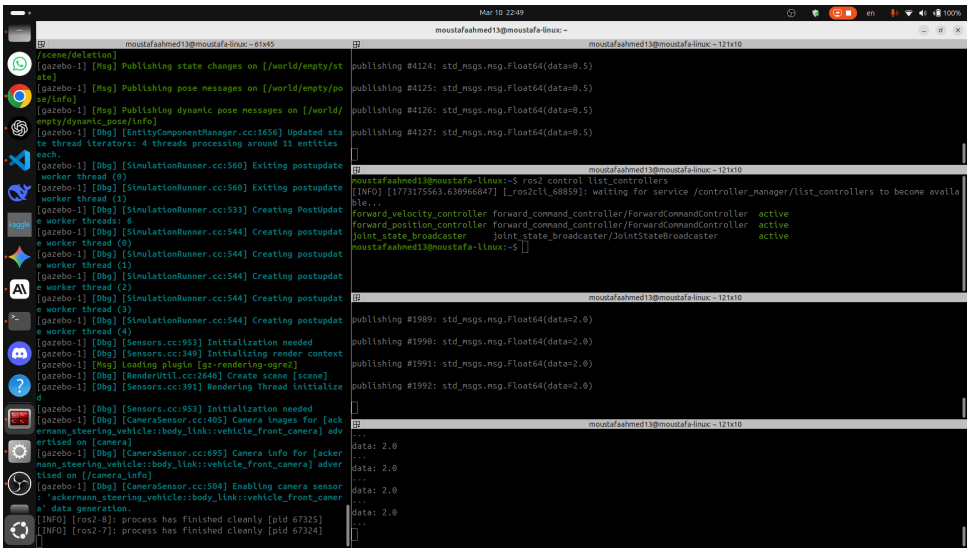
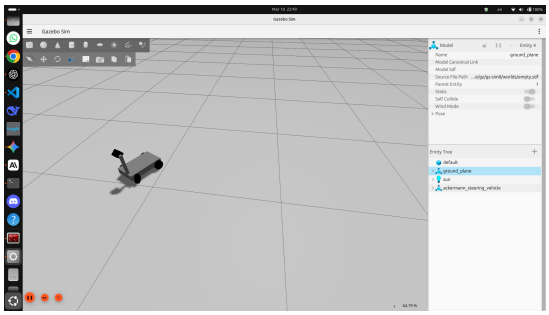


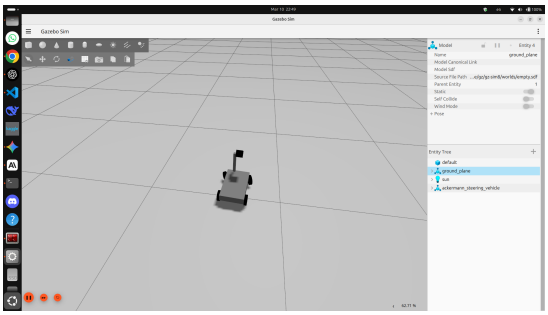
Figure 1: Gazebo and topics

1. Left terminal: running the vehicle model in Gazebo
2. Top right terminal: velocity command
3. Second right terminal: ros2 control list_controllers (all active)
4. Third right terminal: angle command
5. Bottom right terminal: velocity readings

- **Gazebo Screenshots**:



(a) First location



(b) Second location

Figure 2: Vehicle in different locations

- **Validation Node Screenshot**:

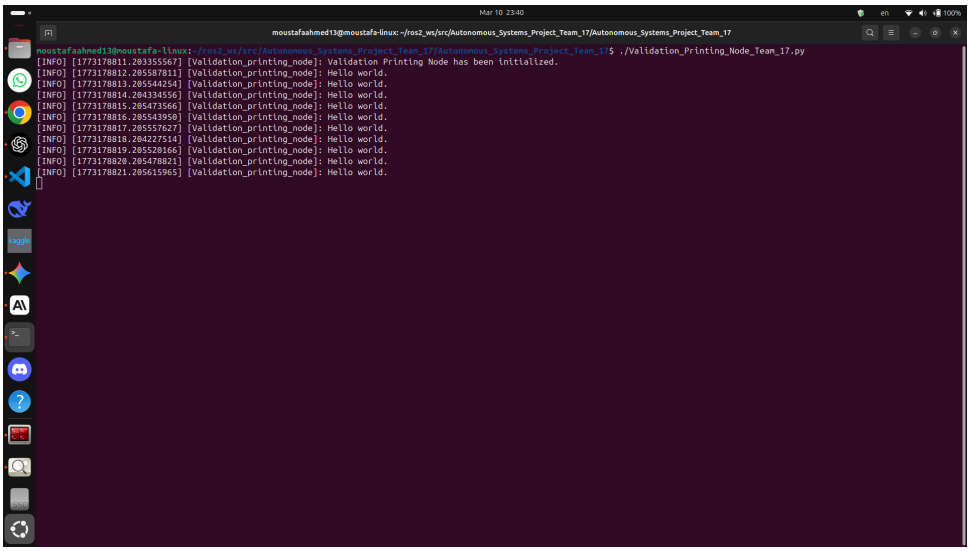


Figure 3: Node printing Hello World

- **Ros2 running**:

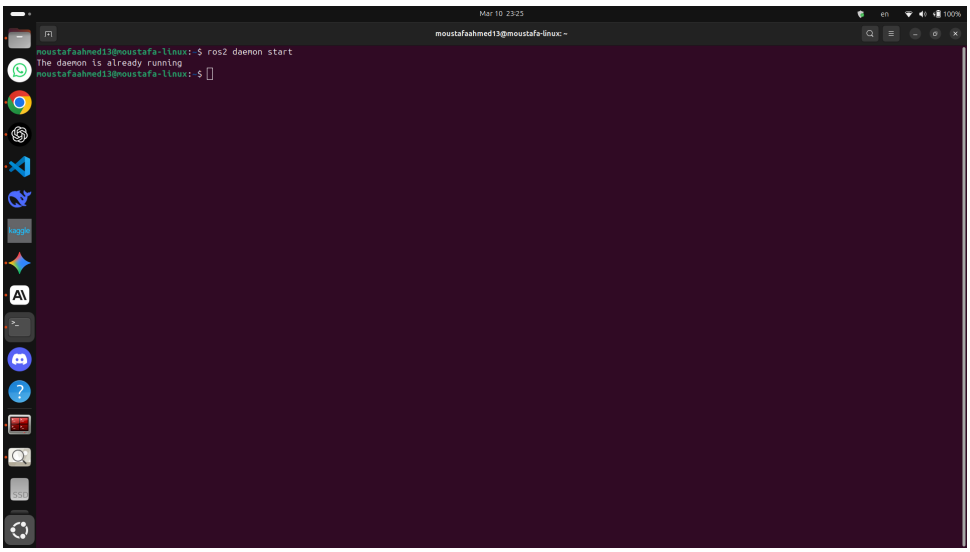


Figure 4: ROS2 running

ROS 2 Topics for Vehicle

- /camera/image_raw *(Subscriber)* – raw camera feed
- /camera/info *(Subscriber)* – camera calibration info
- /clock *(Subscriber)* – simulation clock
- /controller_manager/activity *(Subscriber)* – controller activity
- /controller_manager/introspection_data/full *(Subscriber)* – full introspection data
- /controller_manager/introspection_data/names *(Subscriber)* – introspection data names
- /controller_manager/introspection_data/values *(Subscriber)* – introspection data values
- /controller_manager/statistics/full *(Subscriber)* – full statistics
- /controller_manager/statistics/names *(Subscriber)* – statistics names
- /controller_manager/statistics/values *(Subscriber)* – statistics values
- /diagnostics *(Subscriber)* – system diagnostics
- /dynamic_joint_states *(Subscriber)* – dynamic joint states
- /forward_position_controller/commands *(Publisher)* – position control commands
- /forward_position_controller/transition_event *(Subscriber)* – position controller events
- /forward_velocity_controller/commands *(Publisher)* – velocity control commands
- /forward_velocity_controller/transition_event *(Subscriber)* – velocity controller events

- /joint_state_broadcaster/transition_event *(Subscriber)* – broadcaster events
- /joint_states *(Subscriber)* – joint positions and velocities
- /parameter_events *(Subscriber)* – ROS 2 parameter changes
- /robot_description *(Subscriber)* – URDF robot model
- /rosout *(Subscriber)* – ROS 2 logging output
- /steering_angle *(Publisher)* – steering angle command
- /tf *(Subscriber)* – transform frames
- /tf_static *(Subscriber)* – static transform frames
- /velocity *(Publisher)* – velocity command

Conclusion

The vehicle model is successfully imported into Gazebo, and all relevant ROS 2 topics for controlling and monitoring it are documented. Screenshots are included for further documentation.